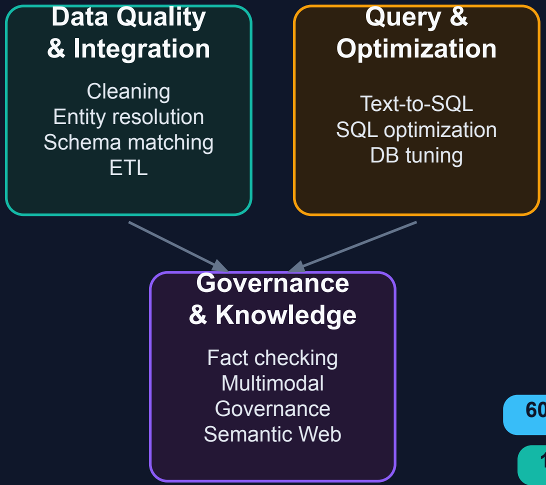
AI Agents for Data Management

A unified synthesis of 60 systems across 11 data-management tasks



60 papers

11 tasks

4 paradigms

From the survey: "A Survey on AI Agents for Data Management: Taxonomy, Progress, and Opportunities"

From scattered prototypes to a coherent field

The survey covers 2023–2025 and keeps only complete end-to-end agentic systems with explicit data-management applications.

60
retained systems in the survey corpus

11
task families summarized

3
functional clusters

4
architectural paradigms

Evolution of the field

The storyline in the survey moves from early autonomous querying to broader enterprise-scale, governance-aware, and explainable operation.

- **2023 — Autonomous querying**: Text-to-SQL, query understanding, basic optimization
- **2024 E — Autonomous DB tuning**: Knob tuning, configuration optimization, online improvement
- **2024 M — Collaborative data prep**: Cleaning, ER, multi-step workflow coordination
- **2024 L — Interactive query refinement**: Multi-turn correction, error recovery, interpretable traces
- **2025 E — Schema-level reasoning**: Schema matching, semantic alignment, explainable support
- **2025 M — Adaptive & explainable operation**: Adaptive planning, symbolic reasoning, production-oriented execution
- **2025 L — Enterprise-scale management**: Scalable orchestration, fault tolerance, governance-aware automation

One taxonomy organizes the whole literature

All surveyed papers are unified first by task: three clusters and eleven operational problem classes.

Data Quality & Integration

Data cleaning
Data Interpreter, DS-Agent, CleanAgent, AutoKaggle

Entity resolution
Structured Multi-Step, Multi-Agent RAG, Harmonia

Schema matching
Agent-OM, LLMA

ETL
DocETL, IntelligentETL

Query & Optimization

Text-to-SQL
ExeSQL, MAC-SQL, SQLFixAgent, MageSQL, Intelli-Dispatch-SQL

SQL optimization
QUITE, GenDB

Database tuning
Panda, AskDB, AgentTune, AMAZe

Governance & Knowledge

Fact checking
FactAgent, LoCal, FIRE, MARO, DelphiAgent, MAD-Sherlock

Multi-modal data
XMODE, M2EX, DataCross, HM-RAG, LightVA

Data governance
DataGovAgent

Semantic Web & KG
KG-Agent, GR-Agent, AgentNet, ChattyKG, KARMA, AGENTiGraph

Names shown are representative systems from the taxonomy figure; the synthesis on the remaining slides abstracts patterns across the full 60-paper corpus.

Beneath the task diversity, the same agent blueprint repeats

1. **Interpret goal** — Natural language intent + schema/data context
2. **Plan** — Break the task into steps or agent roles
3. **Use tools** — SQL engines, Python, retrievers, validators
4. **Execute** — Run queries, repairs, transforms, extraction
5. **Reflect** — Check outputs, repair, retry, critique, debate
6. **Deliver** — Answer, workflow, recommendation, KG update

Persistent state / memory

- Execution traces
- Conversation/shared state
- Case memory / workflow artifacts

Tool layer

- DBMS / SQL engines
- Python & DS runtimes
- RAG, web, graph, vector tools

Assurance layer

- Validation and execution feedback
- Repair / critique / debate loops
- Human oversight for ambiguous cases

Unifying idea: performance comes from orchestrated workflow design—not from the base LLM alone.

Four paradigms dominate the design space

The deeper unifier is architectural organization: who plans, who acts, who validates, and how state is shared.

Single-agent tool-using loop

One agent alternates between reasoning and acting in a tight execution-aware cycle.

Data Interpreter • DS-Agent • ExeSQL • Panda • KG-Agent

Manager-worker / hierarchical

A coordinator delegates subtasks to specialists and integrates their outputs.

CleanAgent • AutoKaggle • AgentTune • DataGovAgent

Collaborative multi-agent

Peer agents critique, debate, or solve complementary subproblems through shared state.

MAC-SQL • SQLFixAgent • Multi-Agent RAG ER • TruEDebate

Planner-executor separation

One module plans or rewrites; another executes, validates, or optimizes.

DocETL • QUITE • ETL and optimization pipelines

Across all eleven fields, the survey's broadest empirical split is still simple: iterative single-agent loops versus shared-state multi-agent coordination.

Cross-task map of recurring design choices

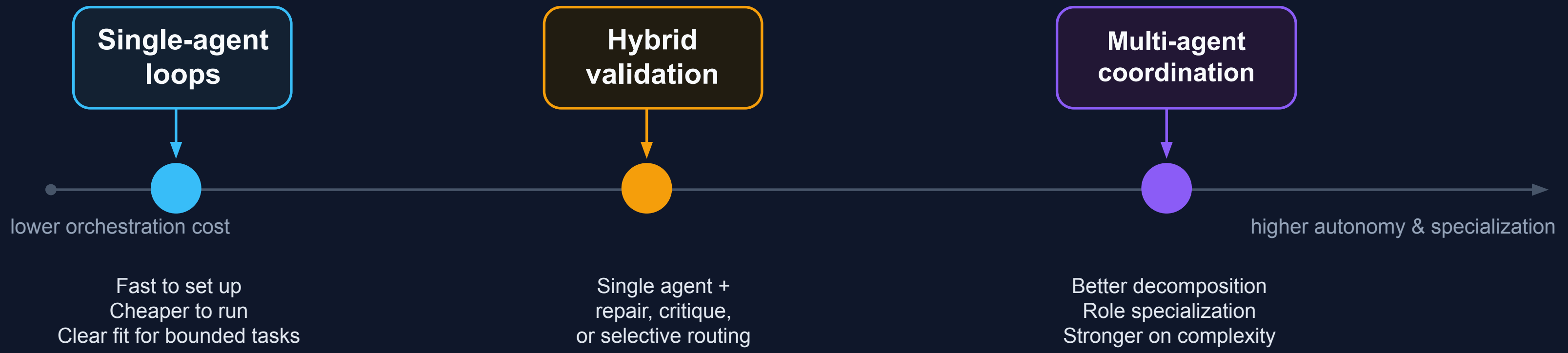
Data modalities and tools change by task; planning, feedback, and coordination patterns recur almost everywhere.

Intensity legend: low | med | high

Task family	Primary data	Tool reliance	Validation loop	Multi-agent
Data cleaning	Tabular	●●●	●●●	●●○
Entity resolution	Structured + docs	●●●	●●○	●●●
Schema matching	Schemas / ontologies	●●●	●●○	●●●
ETL	Mixed enterprise docs	●●●	●●●	●●●
Text-to-SQL	Relational DBs	●●●	●●●	●●●
SQL optimization	Analytical SQL	●●●	●●●	●●●
Database tuning	Workloads + telemetry	●●●	●●●	●●○
Fact checking	Web / multimodal	●●●	●●○	●●●
Multi-modal data	Tables + text + images	●●●	●●○	●●○
Data governance	Enterprise workflows	●●●	●●●	●●○
Semantic Web & KG	Graphs / ontologies	●●●	●●○	●●●

The central trade-off: simplicity vs. autonomy

Single-agent and multi-agent systems appear throughout the survey as complementary design points—not winners in every setting.



Why single-agent persists

- Simpler control flow
- Lower token overhead
- Transparent bounded workflows

Why multi-agent grows

- Complex tasks benefit from decomposition
- Specialists use different tools
- Cross-checking improves robustness

What still blocks production confidence

- No coordination metrics
- Evaluation fragmented by task
- Hallucination / safety risks

What should next generation of researchers focus on

1. **Benchmark the process, not only the final answer** — Reasoning quality, tool-use correctness, collaboration effectiveness, cost, latency, and reliability should all be measured.

2. **Make orchestration robust** — Agents need stronger recovery, safer execution, scalable coordination, and fewer cascading failures.

3. **Fuse symbolic and neural control** — Hybrid architectures can add constraints, traceability, and stronger guarantees to LLM-driven planning.

4. **Generalize across modalities and domains** — Future agents must work across tabular, graph, text, image, and enterprise workflow settings—not in isolated silos.

5. **Keep humans in the loop where stakes are high** — Explainability, calibrated uncertainty, and mixed-initiative collaboration remain essential for governance and trust.